MTH-262 Statistics and Probability Theory

Lecture 15

Discrete Probability Distributions

Introduction and Motivation

No matter whether a discrete probability distribution is represented graphically by a histogram, in tabular form, or by means of a formula, the behavior of a random variable is described.

Often, the observations generated by different statistical experiments have the same general type of behavior.

Consequently, discrete random variables associated with these experiments can be described by essentially the same probability distribution and therefore can be represented by a single formula.

Discrete Probability Distributions

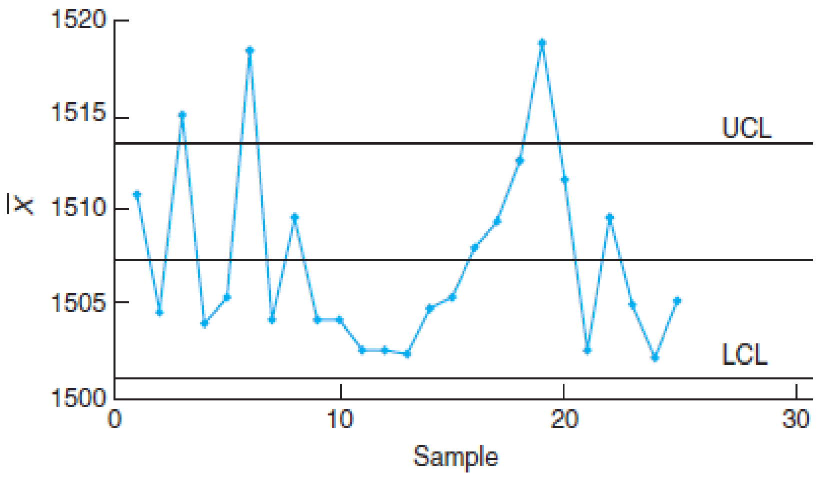
Introduction and Motivation

In fact, one needs only a handful of important probability distributions to describe many of the discrete random variables encountered in practice.

Such a handful of distributions describe several real-life random phenomena. For instance, in a study involving testing the effectiveness of a new drug, the number of cured patients among all the patients who use the drug approximately follows a binomial distribution

In an industrial example, when a sample of items selected from a batch of production is tested, the number of defective items in the sample usually can be modeled as a hypergeometric random variable.

Discrete Probability Distributions



Introduction and Motivation

In a statistical quality control problem, the experimenter will signal a shift of the process mean when observational data exceed certain limits. The number of samples required to produce a false alarm follows a geometric distribution which is a special case of the negative binomial distribution

On the other hand, the number of white cells from a fixed amount of an individual’s blood sample is usually random and may be described by a Poisson distribution.

In this lecture, we discuss the Binomial probability distribution and Multinomial probability distribution.

Discrete Probability Distributions

Binomial Probability Distribution

An experiment often consists of repeated trials, each with two possible outcomes that may be labeled success or failure. The most obvious application deals with the testing of items as they come off an assembly line, where each trial may indicate a defective or a non-defective item. We may choose to define either outcome as a success. The process is referred to as a Bernoulli process. Each trial is called a Bernoulli trial.

Observe, for example, if one were drawing cards from a deck, the probabilities for repeated trials change if the cards are not replaced. That is, the probability of selecting a heart on the first draw is 1/4, but on the second draw it is a conditional probability having a value of 13/51 or 12/51, depending on whether a heart appeared on the first draw: this, then, would no longer be considered a set of Bernoulli trials.

Bernoulli Process

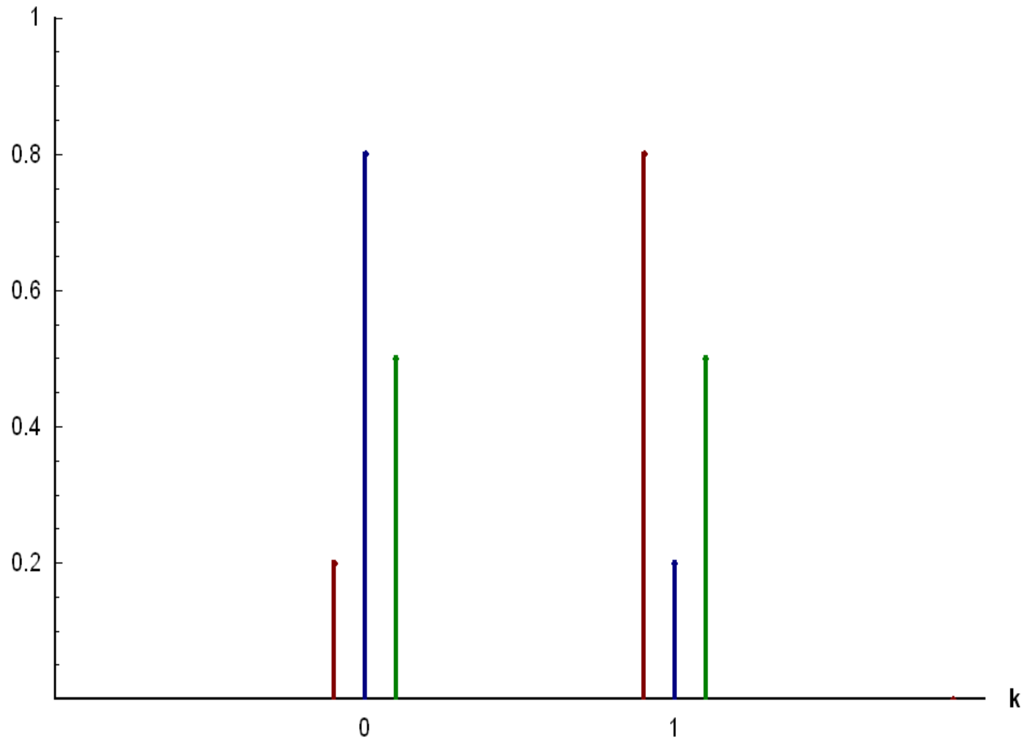
Strictly speaking, the Bernoulli process must possess the following properties:

1. The experiment consists of repeated trials.
2. Each trial results in an outcome that may be classified as a success or a failure.
3. The probability of success, denoted by p, remains constant from trial to trial.
4. The repeated trials are independent.

Probability mass function of Bernoulli Distribution: $p(X=x)=p^x(1-p)^{(1-x)}$ $x=0,1$

Bernoulli Process

- Graph of Bernoulli random variable



----P(X=0)=0.2 and P(X=1)=0.8
----P(X=0)=0.8 and P(X=1)=0.2
----P(X=0)=0.5 and P(X=1)=0.5

Bernoulli Process (Example)

Consider the set of Bernoulli trials where three items are selected at random from a manufacturing process, inspected, and classified as defective or non-defective. A defective item is designated a success. The number of successes is a random variable X assuming integral values from 0 through 3. The eight possible outcomes and the corresponding values of X are:

Outcome	NNN	NDN	NND	DNN	NDD	DND	DDN	DDD
x	0	1	1	1	2	2	2	3

Since the items are selected independently and we assume that the process produces 25% defectives, we have

P(NDN) = P(N)P(D)P(N) =(3/4)*(1/4)*(3/4)

Bernoulli Process (Example)

Since the items are selected independently and we assume that the process produces 25% defectives, we have

$$P(NDN) = P(N)P(D)P(N) = (3/4)*(1/4)*(3/4)$$

Similar calculations yield the probabilities for the other possible outcomes. The probability distribution of X is therefore

x	0	1	2	3
$f(x)$	$\frac{27}{64}$	$\frac{27}{64}$	$\frac{9}{64}$	$\frac{1}{64}$

Binomial Probability Distribution

A Bernoulli trial can result in a success with probability p and a failure with probability q = 1−p. Then the probability distribution of the binomial random variable X, the number of successes in n independent trials, is

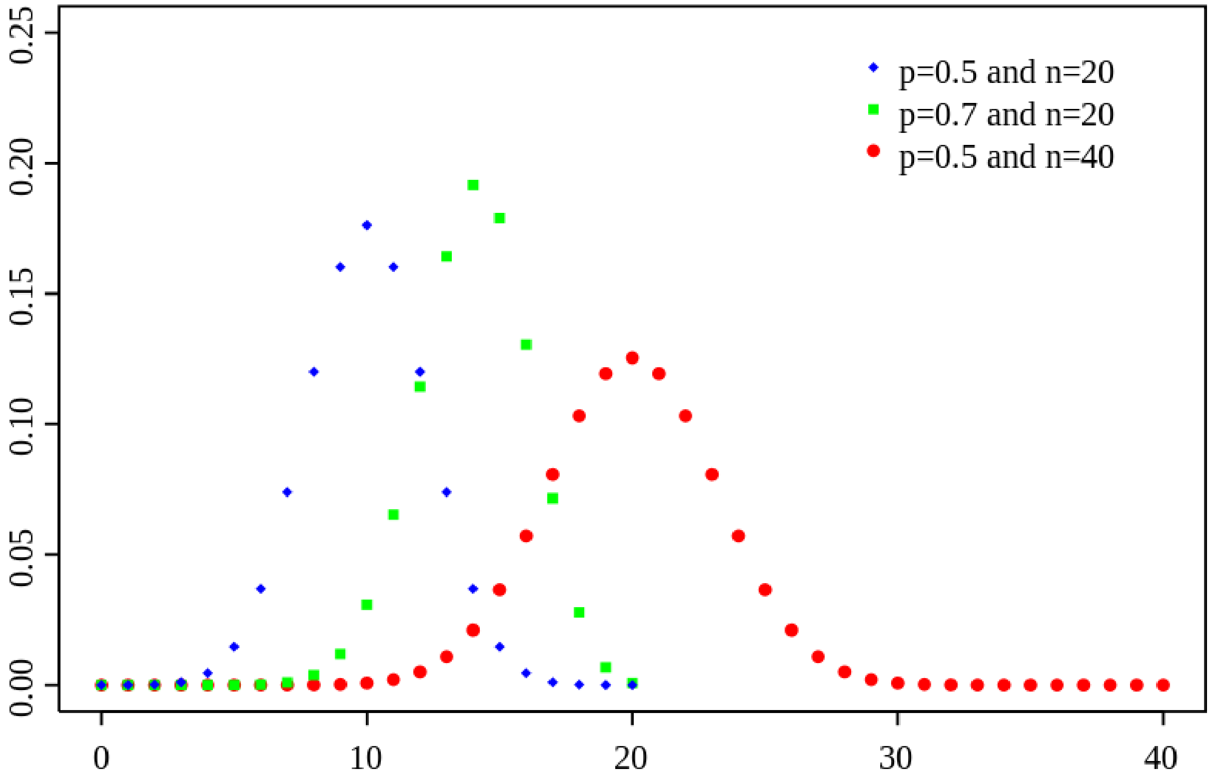
$$b(x; n, p) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \ldots, n.$$

“n” and “p” are the parameters of the Binomial probability distribution.

When $p = 0.5$, the distribution is symmetric around the mean. When $p > 0.5$, the distribution is skewed to the left. When $p < 0.5$, the distribution is skewed to the right.

Graphical Behavior of Binomial Distribution

- When p = 0.5, the distribution is symmetric around the mean. When p > 0.5, the distribution is skewed to the left. When p < 0.5, the distribution is skewed to the right.



Binomial Probability Distribution

Proof of Binomial Distribution:

- Binomial distribution is derived from the rules of probability, product law of probability for independent events.
- For the single trial, the probability distribution is:

X	P(X=x)
0	q
1	P
	(q+p) = 1

$$p(X=x)=p^{x}(1-p)^{(1-x)} \quad x=0,1$$

Binomial Probability Distribution

- For two trials, the probability distribution of a particular random variable is:

X	P(X=x)
0	$q.q=q^2$
1	$q.p+p.q=2.p.q$
2	$p.p=p^2$
Sum	$(q+p)^2=q^2+2.p.q+p^2$

$$p(X=x)=\binom{2}{x}p^x(1-p)^{(2-x)} \quad x=0,1,2$$

Binomial Probability Distribution

- For three trials, the probability distribution of a particular random variable is:

X	P(X=x)
0	$q.q.q=q^3$
1	$q.q.p+q.p.q+p.q.q=3q^2p$
2	$q.p.p+p.q.p+p.p.q=3qp^2$
3	$p.p.p=p^3$
Sum	$(q+p)^3=q^3+3q^2p+3qp^2+p^3$

$$p(X=x)=\binom{3}{x}p^x(1-p)^{(3-x)} \quad x=0,1,2,3$$

Binomial Probability Distribution

- For "n" trials, the probability distribution of a particular random variable is:

X	P(X=x)
0	$q.q.q.....q=q^n$
1	$n.q^{n-1}.p$
.	
.	
.	
n	$p.p.p.....p=p^n$

$$p(X=x)=\binom{n}{x}p^x(1-p)^{(n-x)} \quad x=0,1,2,3,\ldots,n$$

Where Does the Name Binomial Come From?

The binomial distribution derives its name from the fact that the n + 1 terms in the binomial expansion of $(q+p)^n$ correspond to the various values of b(x; n, p) for x = 0, 1, 2, . . . , n. That is,

$$(q+p)^n = \binom{n}{0}q^n + \binom{n}{1}pq^{n-1} + \binom{n}{2}p^2q^{n-2} + \cdots + \binom{n}{n}p^n$$

$$= b(0; n, p) + b(1; n, p) + b(2; n, p) + \cdots + b(n; n, p).$$

Since p + q = 1, we see that

$$\sum_{x=0}^{n} b(x; n, p) = 1,$$

Binomial Probability Distribution (Example)

Question: The probability that a certain kind of component will survive a shock test is 3/4. Find the probability that exactly 2 of the next 4 components tested survive.

Solution: Assuming that the tests are independent and p = 3/4 for each of the 4 tests, we obtain

$$b\left(2;4,\frac{3}{4}\right) = \binom{4}{2}\left(\frac{3}{4}\right)^2\left(\frac{1}{4}\right)^2 = \left(\frac{4!}{2!\,2!}\right)\left(\frac{3^2}{4^4}\right) = \frac{27}{128}.$$

rather than in the tabular form (as written in previous example)

Binomial Probability Distribution (Example)

Question: probability that a patient recovers from a rare blood disease is 0.4. If 15 people are known to have contracted this disease, what is the probability that (a) at least 10 survive, (b) from 3 to 8 survive, and (c) exactly 5 survive?

Solution: Given information:

Variable of Interest = Number of patients survive = X

Number of trials = $n = 15$

Probability of success = $p = 0.4$

Binomial Probability Distribution (Example)

Solution: Given information:

Variable of Interest = Number of patients survive = X

Number of trials = n = 15

Probability of success = p = 0.4

a) $P(X \geq 10) = 1 - P(X < 10) = 1 - \sum_{x=0}^{9} b(x; 15{,}0.4) = 0.0338$

b) $P(3 \leq X \leq 8) = \sum_{x=0}^{8} b(x; 15{,}0.4) - \sum_{x=0}^{2} b(x; 15{,}0.4) = 0.8779$

c) $P(X=5) = ?$

Mean and Variance of the Binomial Probability Distribution

Mean:

$$\mu = np = E(X)$$

$$= \sum_{x=0}^{n} x.\, b(x; n, p)$$

Variance:

$$\sigma^2 = npq = E(X^2) - [E(X)]^2$$

$$= \sum_{x=0}^{n} x^2.\, b(x; n, p) - [\sum_{x=0}^{n} x.\, b(x; n, p)]^2$$

Derivation of Mean

- Mean of the Binomial distribution can be derived as:

$$p(x) = \binom{n}{x} p^x q^{n-x}\text{. Then } E(X) = \sum_{x=0}^{n} x \binom{n}{x} p^x q^{n-x}$$

$$= \sum_{x=0}^{n} x \frac{n!}{(n-x)!x!} p^x q^{n-x}$$

$$= \sum_{x=1}^{n} \frac{n!}{(x-1)!(n-x)!} p^x q^{n-x}$$

(Since the $x = 0$ term vanishes)

$$E(X) = \sum_{x=1}^{n} \frac{n(n-1)!}{(x-1)!(n-x)!} (p) p^{x-1} q^{n-x}$$

$$= np \sum_{x=1}^{n} \frac{(n-1)!}{(x-1)!(n-x)!} p^{x-1} q^{n-x}$$

$$= np \sum_{x=1}^{n} \frac{(n-1)!}{(x-1)!\left[(n-1)-(x-1)\right]!} p^{x-1} q^{n-x}$$

$$= np \sum_{x=1}^{n} \binom{n-1}{x-1} p^{x-1} q^{n-x}$$

$$= np\left[{}^{n-1}C_0 q^{n-1} + {}^{n-1}C_1 p q^{n-2} + {}^{n-1}C_2 p^2 q^{n-3} + \ldots + {}^{n-1}C_{n-1} p^{n-1}\right]$$

$= np$ [This is a binomial expansion of $(p+q)^{n-1}$]

Derivation of Variance

- The variance of the Binomial distribution is:

$$Var[X] = E[X^2] - (E[X])^2$$

$$Var[X] = np^2(n-1) + np - (np)^2$$

$$Var[X] = n^2p^2 - np^2 + np - n^2p^2$$

$$Var[X] = -np^2 + np$$

$$Var[X] = np(-p+1)$$

$$Var[X] = np(1-p)$$

Multinomial Probability Distribution

If a given trial can result in the k outcomes $E_1, E_2, \ldots, E_k$ with probabilities $p_1, p_2, \ldots, p_k$, then the probability distribution of the random variables $X_1, X_2, \ldots, X_k$, representing the number of occurrences for $E_1, E_2, \ldots, E_k$ in n independent trials, is

$$f(x_1, x_2, \ldots, x_k; p_1, p_2, \ldots, p_k, n) = \binom{n}{x_1, x_2, \ldots, x_k} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k},$$

with

$$\sum_{i=1}^{k} x_i = n \text{ and } \sum_{i=1}^{k} p_i = 1.$$

Multinomial Probability Distribution (Example)

Question: The complexity of arrivals and departures of planes at an airport is such that computer simulation is often used to model the "ideal" conditions. For a certain airport with three runways, it is known that in the ideal setting the following are the probabilities that the individual runways are accessed by a randomly arriving commercial jet:

Runway 1: $p_1 = 2/9$, Runway 2: $p_2 = 1/6$, Runway 3: $p_3 = 11/18$.

What is the probability that 6 randomly arriving airplanes are distributed in the following fashion?

Runway 1: 2 airplanes, Runway 2: 1 airplane, Runway 3: 3 airplanes

Multinomial Probability Distribution (Example)

Solution: Given Information:

Runway 1: $p_1 = 2/9$, Runway 2: $p_2 = 1/6$, Runway 3: $p_3 = 11/18$.

Runway 1: 2 airplanes, Runway 2: 1 airplane, Runway 3: 3 airplanes

The probability mass function of the multinomial distribution is

$$f(x_1, x_2, \ldots, x_k; p_1, p_2, \ldots, p_k, n) = \binom{n}{x_1, x_2, \ldots, x_k} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k},$$

$$f\left(2, 1, 3; \frac{2}{9}, \frac{1}{6}, \frac{11}{18}, 6\right) = \binom{6}{2, 1, 3} \left(\frac{2}{9}\right)^2 \left(\frac{1}{6}\right)^1 \left(\frac{11}{18}\right)^3$$

$$= \frac{6!}{2!\,1!\,3!} \cdot \frac{2^2}{9^2} \cdot \frac{1}{6} \cdot \frac{11^3}{18^3} = 0.1127.$$

Question

- A manufacturer knows that on average 20% of the electric toasters produced require repairs within 1 year after they are sold. When 20 toasters are randomly selected, find appropriate numbers x and y such that
- (a) the probability that at least x of them will require repairs is less than 0.5;
- (b) the probability that at least y of them will not require repairs is greater than 0.8.

Binomial Probabilities Table

20	**0**	0.1216	0.0115	0.0032	0.0008	0.0000					
	1	0.3917	0.0692	0.0243	0.0076	0.0005	0.0000				
	2	0.6769	0.2061	0.0913	0.0355	0.0036	0.0002				
	3	0.8670	0.4114	0.2252	0.1071	0.0160	0.0013	0.0000			
	4	0.9568	0.6296	0.4148	0.2375	0.0510	0.0059	0.0003			
	5	0.9887	0.8042	0.6172	0.4164	0.1256	0.0207	0.0016	0.0000		
	6	0.9976	0.9133	0.7858	0.6080	0.2500	0.0577	0.0065	0.0003		
	7	0.9996	0.9679	0.8982	0.7723	0.4159	0.1316	0.0210	0.0013	0.0000	
	8	0.9999	0.9900	0.9591	0.8867	0.5956	0.2517	0.0565	0.0051	0.0001	
	9	1.0000	0.9974	0.9861	0.9520	0.7553	0.4119	0.1275	0.0171	0.0006	
	10		0.9994	0.9961	0.9829	0.8725	0.5881	0.2447	0.0480	0.0026	0.0000
	11		0.9999	0.9991	0.9949	0.9435	0.7483	0.4044	0.1133	0.0100	0.0001
	12		1.0000	0.9998	0.9987	0.9790	0.8684	0.5841	0.2277	0.0321	0.0004
	13			1.0000	0.9997	0.9935	0.9423	0.7500	0.3920	0.0867	0.0024
	14				1.0000	0.9984	0.9793	0.8744	0.5836	0.1958	0.0113
	15					0.9997	0.9941	0.9490	0.7625	0.3704	0.0432
	16					1.0000	0.9987	0.9840	0.8929	0.5886	0.1330
	17						0.9998	0.9964	0.9645	0.7939	0.3231
	18						1.0000	0.9995	0.9924	0.9308	0.6083
	19							1.0000	0.9992	0.9885	0.8784
	20								1.0000	1.0000	1.0000

Binomial Distribution in R

- In R, the sample command can be used to generate random samples from this distribution. For example, to generate ten random samples when p =1/4 can be done with

```
n <- 10; p <- 1/4
sample(0:1, size=n, replace=TRUE, prob=c(1-p, p))

##  [1] 0 0 0 1 0 1 0 0 0 0
```

Binomial Distribution in R

- Example 6.8: Tossing ten coins Toss a coin ten times. Let X be the number of heads. If the coin is fair, X has a Binomial(10,1/2) distribution.
- The probability that X = 5 can be found directly from the distribution with the choose function:

```
choose(10,5) * (1/2)^5 * (1/2)^(10-5)

## [1] 0.2461
```

```
dbinom(5, size=10, prob=1/2)

## [1] 0.2461
```

Binomial Distribution in R

- The probability that there are six or fewer heads, $P(X \leq 6)$ can be given either of these two ways:

```
sum(dbinom(0:6, size=10, prob=1/2))

## [1] 0.8281

pbinom(6, size=10, p=1/2)

## [1] 0.8281
```

Binomial Distribution in R

- If we wanted the probability of seven or more heads, we could answer using $P(X \geq 7) = 1 - P(X \leq 6)$, or using the extra argument lower.tail=FALSE. This returns $P(X > k)$ rather than $P(X \leq k)$:

```
sum(dbinom(7:10,size=10,prob=1/2))

## [1] 0.1719

1 - pbinom(6,size=10,p=1/2)

## [1] 0.1719

pbinom(6,size=10,p=1/2, lower.tail=FALSE) # k = 6 not 7!

## [1] 0.1719
```